

The background is a blurred image of a computer monitor. On the left, a portion of a Windows calculator is visible, showing a grid of buttons. The main area of the screen displays lines of code in a dark-themed editor, with some text highlighted in green. At the bottom, a Windows taskbar is visible with various application icons.

C#

Level 1

Sum of Matrix



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Definition

- ◆ An implicitly typed variable is a local variable whose type is automatically determined by the compiler using the `var` keyword.
- ◆ You do not explicitly write the data type; `C#` determines it from the value assigned to the variable.

Explanation

◆ Instead of writing:

```
int x = 10;  
double y = 10.5;  
string z = "Sara";
```

◆ You can write:

```
var x = 10;  
var y = 10.5;  
var z = "Sara";
```

 $x \rightarrow \text{int}$

 $y \rightarrow \text{double}$

 $z \rightarrow \text{string}$

Procedures

1. Declare the Variable

-  Use the var keyword:

var x = 10;

2. Initialize the Variable

- ◆ A variable declared with var must be initialized at the time of declaration.

var x = 10; // ✓

var name = "Sara"; // ✓

3. Compiler Determines the Type

var x = 10;

var y = 10.5;

var z = "Sara";

 x → int

 y → double

 z → string

💡 Example

```
var x = 10;  
var y = 10.5;  
var z = "Sara";  
Console.WriteLine("x={0}, y={1}, z={2}", x, y, z);  
Console.ReadKey();
```



x becomes an int.
y becomes a double.
z becomes a string.

Note

- ◆ var is not dynamic typing.
var x = 10;
- ◆ After compilation, x is an int. You cannot later do:
x = "Sara"; // ✗ Error

⊘ Warning

⚠ You cannot declare a var variable without assigning a value:

`var x; // ✗ Error`

⚠ var does not mean "any type."

⚠ The type is determined at compile time, not while the program is running.



Comparison

▨ Explicit Type

- ✓ The programmer specifies the type.
- ✓ The variable can be declared with a specific type.
- ✓ The type is visible directly in the declaration.

▤ Implicit Type

- ✓ The compiler determines the type from the assigned value.
- ✓ var must be initialized when declared.
- ✓ The variable still has a specific type after compilation

✦✦ Summary

- ✓ `var` is used to declare implicitly typed local variables.
- ✓ The compiler automatically determines the variable's type.
- ✓ The variable must be initialized when declared.
- ✓ `var x = 10;` → `x` is `int`.
- ✓ `var y = 10.5;` → `y` is `double`.
- ✓ `var z = "Sara";` → `z` is `string`.
- ✓ `var` is still strongly typed; it does not mean "any type."

Thank you 😊